# Realistic Agent-based Simulation of Evolutionary Game Theory

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**Abstract:** Evolutionary Game Theory (EGT) provides a fundamental framework for understanding population dynamics and strategic interactions in biological systems. However, traditional mathematical models often struggle to capture the complexities inherent in real-world evolutionary processes. In this research, we propose a novel agent-based simulation tool aimed at addressing these limitations. By incorporating diverse initial trait distributions, dynamic environmental pressures, spatial interactions, and population size changes, our simulation offers a comprehensive and realistic framework for studying evolutionary dynamics. Through simulations, we observe significant impacts on evolutionary outcomes, such as the emergence and persistence of adaptive strategies, the influence of environmental factors like temperature fluctuations, and the role of spatial interactions in shaping population dynamics. Our model represents a valuable contribution to the field of evolutionary dynamics, providing researchers with a powerful platform for exploring complex ecological interactions and evolutionary processes in biological populations.

*Keywords:* Evolutionary Game Theory, Agent-based Simulation, Population Dynamics, Evolutionary Adaptation

## 1. INTRODUCTION AND RESEARCH MOTIVATION

Evolutionary Game Theory (EGT) stands as a fundamental framework for understanding population dynamics and strategic interactions in biological systems. However, traditional mathematical models in EGT often struggle to capture the complexities inherent in real-world evolutionary processes, such as diverse initial trait distributions, dynamic environmental pressures, and a lack of empirical validation [1][2]. This limitation has been underscored by significant contributions from related works, such as "Evolutionary game theory using agent-based methods" [3], which highlight the shortcomings of traditional models and advocate for the adoption of agent-based simulations to capture evolving populations more realistically [4].

Despite these advancements, gaps persist in the current literature, particularly in the limited consideration of initial diversity in strategies, complex environmental dynamics, neglect of spatial interactions, and dynamic population sizes [3]. This proposal of a novel research simulation tool aims to address these gaps by leveraging agent-based simulations, providing a more adaptable and realistic framework for studying evolutionary dynamics. By incorporating diverse initial trait distributions, dynamic environmental pressures, spatial interactions, and accounting for population size changes, this tool seeks to create a comprehensive simulation model that better mirrors real-world evolutionary processes.

By allowing for the input of a diverse range of strategies by the researcher, this research tool introduces a novel departure from homogeneous initial trait distributions. The inclusion of environmental pressures such as temperature, exerts selective pressures on populations, influencing the evolution of strategies. Considering spatial interactions where agents move around and interact only with nearby agents, introduces a realistic aspect to the simulation [4]. Lastly, incorporating birth and death rates to simulate growing and declining population sizes adds a stochastic element to the model, reflecting the natural fluctuations in population sizes observed in real-world scenarios.

Overall, the proposed novel approach represents a significant advancement in the study of evolutionary dynamics, offering a more comprehensive and realistic framework for understanding how biological populations adapt and evolve.

## 2. PROBLEM STATEMENT AND OBJECTIVES

### *2.1 Problem Statement*

"Can the development of a more realistic agent-based simulation tool, incorporating diverse initial trait distributions, dynamic environmental pressures, spatial interactions, and population size changes, facilitate a better understanding of evolutionary dynamics and adaptation in biological populations?"

### 2.2 Objectives

- Develop a comprehensive agent-based simulation tool using Python to model evolutionary dynamics more realistically among animal populations.
- Investigate the effects of diverse initial trait distributions on the emergence and persistence of adaptive strategies within populations.
- Explore the influence of dynamic environmental pressures, such as temperature fluctuations, on the evolution of strategies and behaviors.
- Analyze the role of spatial interactions - such as movement, competition and cooperation patterns - in shaping population dynamics and species coexistence [4].
- Examine the impact of birth and death rates on population sizes and assess their implications for long-term evolutionary trajectories.
- Generate interactive visualizations and data analysis tools within the simulation environment to facilitate interpretation and exploration of evolutionary dynamics.

## 3. LITERATURE REVIEW

Research on mathematical evolutionary game theory (EGT) and adaptive dynamics has contributed significantly to understanding how gradual evolution leads to adaptation in populations with interacting individuals [1][2]. This body of work has provided essential mathematical tools and models to study trait evolution, including frequency-dependent and density-dependent selection processes that drive adaptation and the emergence of polymorphism. However, past research in this area has faced criticism for its limited consideration of initial trait diversity, lack of empirical validation for mathematical models, oversimplified environmental dynamics, and neglect of spatial complexities in species interactions.

The paper “Evolutionary game theory using agent-based methods” [2] attempted to address these limitations. This paper provides a robust framework to understand the evolution of strategies in populations facing conflicts. While traditional mathematical models are effective in predicting optimal strategies in simple scenarios, agent-based methods are essential for more realistic settings involving finite populations, mutations, stochastic decisions, and communication among agents. This paper emphasizes the limitations of mathematical treatments and highlights the need for agent-based simulations to capture the complexities of evolutionary outcomes.

### 3.1 Strengths

- The paper provides a comprehensive analysis of the limitations of traditional mathematical game theory models when applied to realistic evolutionary scenarios involving finite populations, mutations, stochastic decisions, and communication between agents.
- By incorporating agent-based simulations, the research extends beyond the constraints of mathematical equations to simulate the complexities of evolving populations with heterogeneous strategies and changing interactors over time.
- The discussion on mutation rates and population sizes in relation to evolutionary dynamics provides empirical relevance, especially in biological contexts.

### 3.2 Weaknesses/Research Gaps

- Lack of diversity in strategies: The paper’s limited consideration of initial strategies overlooks real-world complexities and diversity.
- Complex Environmental Dynamics: Failure to capture dynamic environmental factors hinders accurate trait adaptation modeling, impacting real-world applicability.
- Unrealistic Spatial Interactions: Although the paper speaks about spatial interactions, inadequate incorporation of spatial interactions limits the model’s realism.
- Constant population size: Although the model assumes a finite population size similar to real-world scenarios, it does not account for growing/declining population sizes.

### 3.3 Addressing the Research Gaps

- Incorporating a diverse range of strategies representing real-world traits by letting the researcher input the strategies.
- Including environmental pressures such as temperature fluctuations, which are crucial factors influencing evolutionary dynamics.
- Agents interact only with nearby agents, simulating realistic spatial dynamics.
- Incorporating birth and death rates, to simulate growing/declining population sizes.

## 4. METHODOLOGY

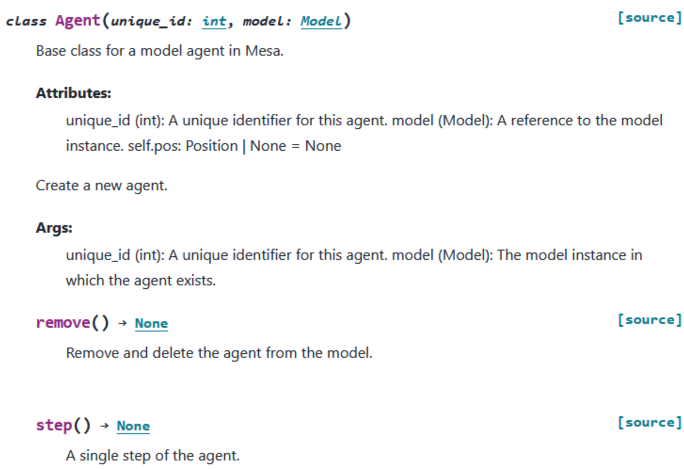
(Link to our Code)

The methodology involves developing a realistic agent-based simulation model using Python programming language [3][4][5]. This sample model will simulate evolutionary dynamics among animal populations, focusing on strategies, interactions, and pay-off matrices that influence adaptation and survival.

Our methodology includes the following terms incorporated in the tool:

(1) **Agents:**
Agents represent individual animals and are implemented as a class in Python [4]. Each agent object encapsulates its strategies, pay-off matrices, spatial interactions, survival probability, and reproduction rate.
In our example model, we have taken 4 agents from the Arctic region - arctic hare, white arctic wolf, grey wolf and polar bear. The researcher can input various other diverse agents on their own.

(2) **Strategies:**
Strategies define how species adapt to their environment, such as coat color and fur thickness for arctic wolves [3]. Researchers input various strategies from a study, which are combinations of observed traits like (white coat, thick fur) or (grey coat, thick fur).



```
class Agent(unique_id: int, model: Model)                                   [source]
    Base class for a model agent in Mesa.

    Attributes:
        unique_id (int): A unique identifier for this agent. model (Model): A reference to the model
        instance. self.pos: Position | None = None

    Create a new agent.

    Args:
        unique_id (int): A unique identifier for this agent. model (Model): The model instance in
        which the agent exists.

    remove() → None                                                         [source]
        Remove and delete the agent from the model.

    step() → None                                                           [source]
        A single step of the agent.
```

Fig. 1. Agent Class

In our example model, we have taken strategies for each of our 4 agents, including information like few phenotypic adaptations, optimal range of temperature for survival, and relative speed of movement compared to other co-agents [6][7][8].

(3) **Pay-off Matrices:**
Pay-off matrices quantify the cost of interactions between an agent's strategies and those of its co-agents [3]. Each entry in the pay-off matrix corresponds to the pay-off received by the animal on the left side of the matrix on interacting with the animal on the top side of that specific matrix element. For example, a polar bear (predator) with a strategy of (short ear, long claws) interacting with a wolf (prey) with (grey coat, thick fur) would have an example pay-off matrix showing the costs and gains of such interactions:

| | **W (grey coat, thick fur)** | **P (short ear, long claw)** |
|---|---|---|
| **W (grey coat, thick fur)** | 0 | -2 |
| **P (short ear, long claw)** | 1 | 0 |

Table 1. Payoff Matrix

The payoff matrix determines an agent's fitness by aggregating the payoffs from all neighboring interactions within a radius in a single time step.
In our example model, we have taken into account the following facts as well [6][7][8]:
- Wolves are a prey for the polar bear. Arctic wolves having white fur color, are able to camouflage, and escape better. Grey wolves, not being able to adapt to the arctic conditions well, are an easier prey. Thus, grey wolves receive a lower pay-off than arctic wolves when interacting with a polar bear. The same way, the polar bear receives a higher pay-off when interacting with a grey wolf.
- Arctic wolves are the best predators of arctic hares, as they well-adapted to the arctic region and rely on hares for a major part of their diet. Thus they receive the highest pay-off when interacting with the hares. Grey wolves and polar bears receive a slightly lower positive pay-off, as they prefer other prey over hares.
- Arctic hares benefit the most when interacting with each other, as they are more protected, receiving the highest pay-off when together. The wolves receive a lower positive pay-off due to their occasional pack mentalities, but occasional aggressiveness as well. Polar bears receive the least positive pay-off as they prefer to be solitary hunters, but occasionally team up together for larger prey.

(4) **Environment:**
Agents are placed on the grid and can move around within the grid boundaries. The interaction radius determines the neighboring cells an agent can interact with.

(5) **Reproduction & Mutations:**
- The reproduction probability is calculated using the agent's relative fitness, scaled between the minimum and maximum fitness values for that strategy.
- If an agent reproduces, a new agent (child) is created and placed in an empty neighboring cell (only if there is one) within the reproduction radius.
- Mutation can occur during reproduction, allowing for the child agent to have a different strategy than the parent.
- In this specific example model, mutation only happens between the arctic wolf and grey wolf strategies, with a certain mutation rate.

(6) **Birth & Death Rates:** Initial population of each agent is specified. The birth rate is implicitly determined by the reproduction mechanism and the availability of empty neighboring cells for new agents to be placed. The death rate is influenced by several factors:
- Lower the fitness of agent, higher the chance of its death.
- A timer mechanism that kills agents after a certain number of steps (e.g., 3 steps), to incorporate the number of generations an agent can survive for.
- The fraction of occupied cells in the surrounding area, representing competition for resources. A higher fraction of occupied cells leads to a higher probability of death.
- Environmental factors such as temperature, also influence the probability of death.

(7) **Movement:**
- Each agent has a movement distance attribute, which determines the radius in which the agent can move in a single step. This replicates the relative speed with which an agent can move, with hares moving faster than the wolves, which in turn move faster than the polar bears, in our example model.
- The movement cell is randomly chosen from within the radius.
- If an agent is unable to find an empty cell within its movement distance, it remains in its current position.

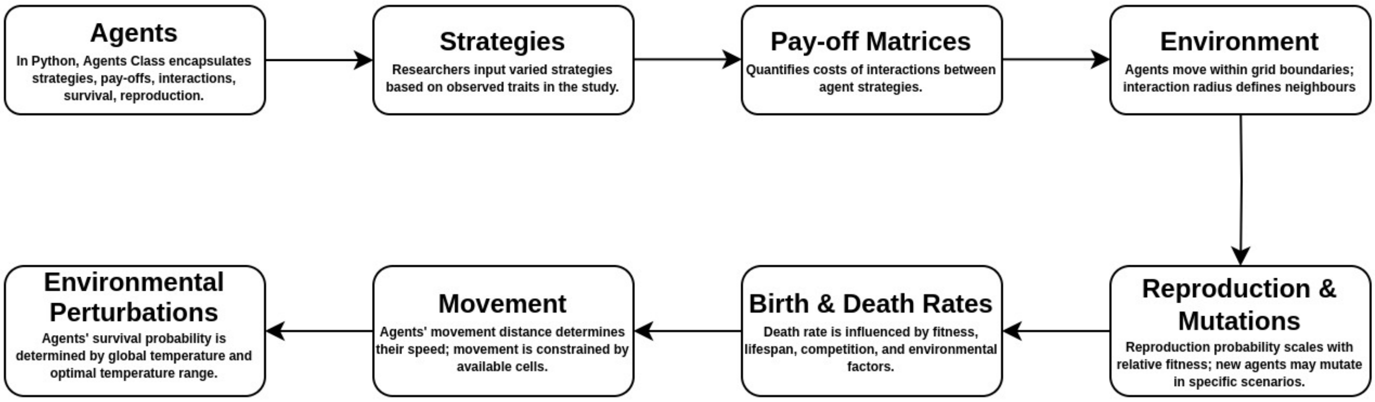



Fig. 2. Methodology Flowchart

- Until a certain time, the child follows the same direction as parent. This replicates nature, how children stay close to their parent until they are grown up.
- Movement enables agents to explore the environment, interact with other agents, and potentially reproduce in different locations.

(8) **Environmental Perturbations:**
We are using temperature as a factor. Each strategy has an optimal temperature range within which it can survive the best. The global temperature in the environment is used to calculate an agent's survival probability using the normal distribution formula:

$$e^{-\left(\frac{\text{global\_temp\_optimal\_temp[0]}}{\text{optimal\_temp[1]}}\right)^2} \qquad (1)$$

where *'global_temp'* is the temperature of environment, *'optimal_temperature[0]'* is the temperature at which the agent will survive with probability 1, and *'optimal_temperature[1]'* shows how sensitive the agent is to a temperature that is not it's optimal temperature. If the global temperature deviates significantly from the optimal range, the agent's survival probability decreases, increasing the likelihood of death.

## 5. RESULTS AND ANALYSIS

In this section, we present the outcomes of our attempt of a realistic agent-based simulation model. The results are organized to highlight key findings relevant to the study's objectives.

(1) **Incorporation of Diverse Initial Trait Distributions:**
We were able to incorporate a range of trait distributions initially, to get a more realistic evolutionary simulation outcome. We successfully included traits such as phenotypic adaptations (like coat colour, fur thickness, ear length and claw length), optimal range of temperature for survival, and relative speed of movement compared to other co-agents. We were also able to successfully simulate the population to predict evolutionary outcomes over a long-term of 100 time steps. We just used a small set of four strategies. However, researchers can input many more diverse initial traits, and as a result study their interactions and long-term outcomes.

(2) **Impact of Environmental Temperature:**
The global temperature of -16°C was chosen to reflect typical arctic conditions. The survival probabilities of the species, based on their optimal temperature ranges, significantly influenced population dynamics. For instance, arctic wolves and polar bears, better adapted to colder temperatures, exhibited higher survival rates.

(3) **Spatial Interactions and Movement Patterns:**
Agents moved within a grid environment, interacting with nearby agents. The movement patterns and spatial interactions had a notable impact on population dynamics:

- Predator-Prey Interactions: Predation events were frequent when predators (wolves and polar bears) encountered prey (wolves and hares). The spatial distribution of these interactions led to areas of high predation pressure, influencing the movement and survival strategies of prey.
- Cooperation and Competition: Arctic hares were almost always found in groups offering protection. Wolves occasionally formed cooperative hunting groups, which improved their hunting success rates. In contrast, polar bears primarily operated as solitary hunters, which influenced their spatial distribution and interaction frequencies.

(4) **Competition for Resources:**
The model incorporated competition for resources by considering the availability of empty cells in the environment. Agents experienced higher death rates when the fraction of occupied cells in their vicinity was high, indicating resource scarcity.

(5) **Reproduction and Mutation Effects:**
Reproduction probabilities were calculated based on agents' fitness, which was influenced by their interactions and survival in the environment. The inclusion of mutation rates allowed for the emergence of new strategies:

- Adaptive Mutations: Occasional mutations between arctic and grey wolves led to new adaptive traits. However, the mutation rate was low, resulting in gradual evolutionary changes rather than abrupt shifts in population dynamics.

- Higher fitness agents had greater reproductive success, leading to the proliferation of adaptive traits over time. The availability of empty neighboring cells also played a crucial role in the reproduction process.

(6) **Birth and Death Rates:**
Birth rates were influenced by factors such as available resources, fitness levels, and environmental conditions. Death rates were influenced by predation, competition, environmental factors, and individual fitness levels.

We used matplotlib to generate visualizations of the grid environment and population dynamics. The following visualizations provide insights into the simulation outcomes:

**Grid Visualization:**
A visual representation of the grid environment showing the spatial distribution of agents at different simulation steps. Agents are color-coded based on their species (yellow - arctic hare, white - arctic wolf, grey - grey wolf, blue - polar bear), providing a clear view of movement patterns and interactions. (Attached is a link of our results as an animation).

A few extra observations from the results' animation were:

- Initially the population was scattered, but they soon huddle up in groups according to each agents' convenience; except polar bears, as they prefer to be solitary hunters.
- The wolves seem to be moving towards the hares, to move closer to their prey as it is advantageous.
- The hares then try to move away, but the wolves keep following behind.
- The polar bears prefer the wolves over the hares as prey, although both are at reachable distance.
- Polar bears are moving towards their prey, but slower as their relative speed is slower due to their big size.

**Strategy Distribution Plot:**
A line plot illustrating the changes in population sizes of each species over time (Fig. 3). This plot helps identify trends in population dynamics and the effects of different factors on species survival.

## 6. LIMITATIONS OF THE MODEL

- **Constant Number of Children:** The model assumes a constant number of children per agent, potentially oversimplifying the variability seen in real-world reproductive patterns.
- **Single-Time Step Reproduction:** Reproduction occurs within a single time step, which may not fully capture the complexities of natural reproductive cycles and intervals.
- **Resource Independent Survival:** While agents with better access to food and resources receive higher fitness scores, the model allows for survival and reproduction even in the absence of these resources, which may not fully capture the impact of resource availability on natural selection.
- **Fixed Temperature Assumption:** The model maintains a constant temperature, neglecting the impact of temperature fluctuations on species behavior and evolutionary adaptations.

## 7. WORK DONE AFTER PROJECT TALK

(Here is the updated code) After the project talk, we identified and addressed few areas to enhance the realism and functionality of our simulation model. One key limitation was the fixed temperature assumption, which did not account for the natural fluctuations in environmental conditions that affect evolutionary dynamics. To address this, we introduced a mechanism to simulate random temperature fluctuations. Specifically, our model now generates a random decimal number between -3 and 3 and adds this value to the current global environment temperature for the next generation. This addition introduces variability and mimics more realistic temperature changes. Future improvements could involve more targeted temperature fluctuations, such as those related to global warming scenarios. Fig. 4 shows the updated strategy distribution line plot. Additionally, we enhanced our visualization model to provide a more comprehensive view of the simulation's progress and environmental conditions. The updated visualization includes an animation of the grid environment, showing the spatial distribution and movement of agents over time. Alongside this animation, the current temperature is displayed on the side, allowing for real-time observation of how temperature fluctuations impact the population dynamics and agent behaviors. (Attached is the same link of our new results as an animation).

The enhanced visualization tool aids in better understanding and interpreting the complex interactions and adaptive strategies within the simulated ecosystem.

## 8. FUTURE DIRECTIONS

There are several areas for potential future exploration and development. One key area is the incorporation of variable reproductive rates and intervals to better mimic real-world reproductive dynamics. This could provide a more accurate representation of species propagation. Additionally, the development of a more nuanced reproductive model that accounts for seasonal or cyclic reproductive patterns could offer further insights into the complex dynamics of species reproduction. Finally, integrating resource-based survival mechanisms could reflect the influence of resource availability on species fitness and survival. This would allow for a more comprehensive understanding of how environmental factors can impact species populations. These enhancements could significantly improve the accuracy and applicability of our current models.

## 9. CONCLUSION

In conclusion, our research presents a novel agent-based simulation tool designed to address the limitations of traditional mathematical models in Evolutionary Game Theory and pre-existig agent-based models. By incorporating diverse initial trait distributions, dynamic environmental pressures, spatial interactions, and population size changes, our simulation offers a comprehensive and realistic foundational framework for studying evolutionary

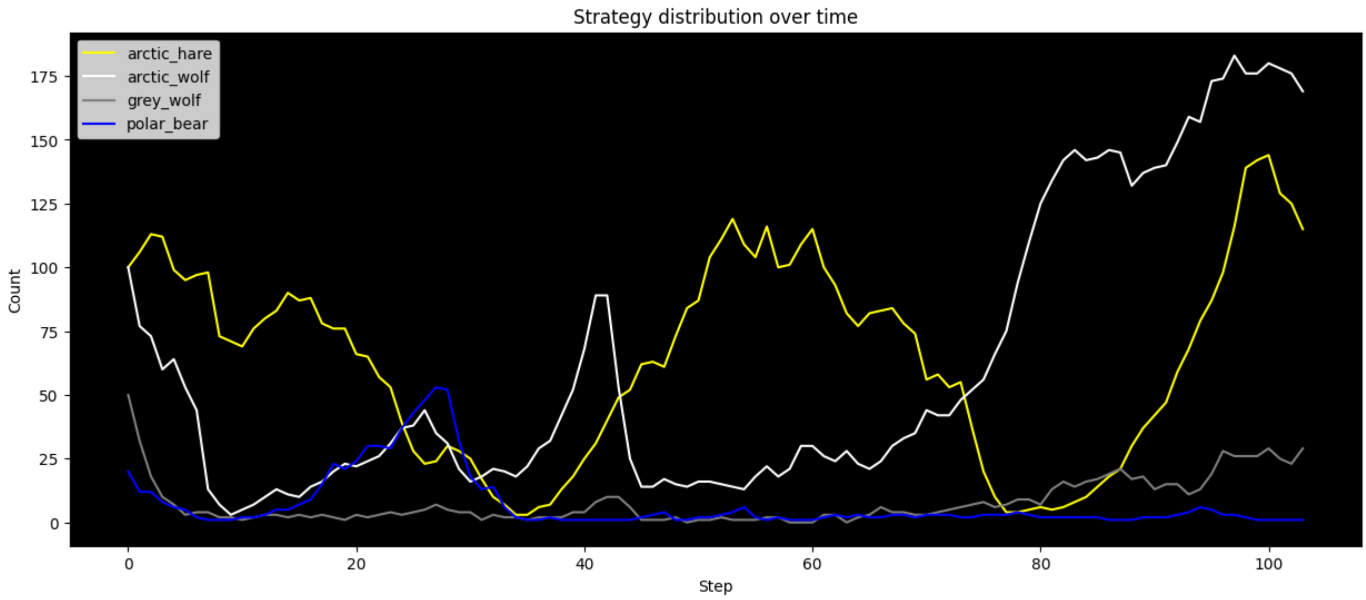



Fig. 3. Strategy Distribution Plot over Simulation of 100 Time Steps. It is seen that the population of hares fluctuates a lot due to predation pressure and re-location. Due to a low population size of polar bears, it remains pretty much constant, but spikes up in between due to proximity to prey. Similar trends are seen for arctic and grey wolves due to predation pressure and prey availability.

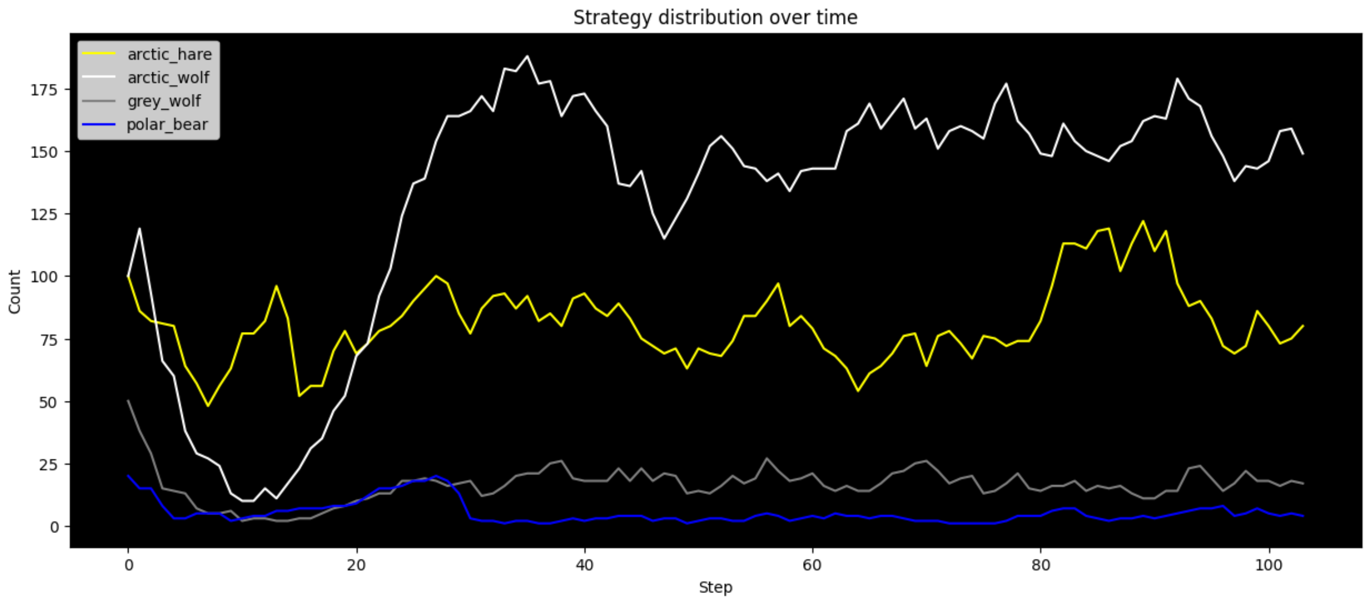



Fig. 4. The Updated Strategy Distribution Plot over Simulation of 100 Time Steps with Temperature Fluctuations.

dynamics.

Through our simulation, we observed significant impacts on evolutionary outcomes, such as the emergence and persistence of adaptive strategies, the influence of environmental factors like temperature fluctuations, and the role of spatial interactions and varying population sizes in shaping population dynamics and species coexistence. These insights contribute to a deeper understanding of how biological populations adapt and evolve in response to their environment.

While our model has certain limitations, such as simplifications in reproductive patterns and resource dependency, it does serve as a very strong foundational and novel framework. This will further open doors to future enhancements to further refine the accuracy and applicability of our simulation.

Overall, our agent-based simulation tool represents a valuable contribution to the field of evolutionary dynamics, offering researchers a powerful platform for exploring complex ecological interactions and evolutionary processes in biological populations.

## 10. ACKNOWLEDGEMENTS AND AUTHOR CONTRIBUTIONS

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